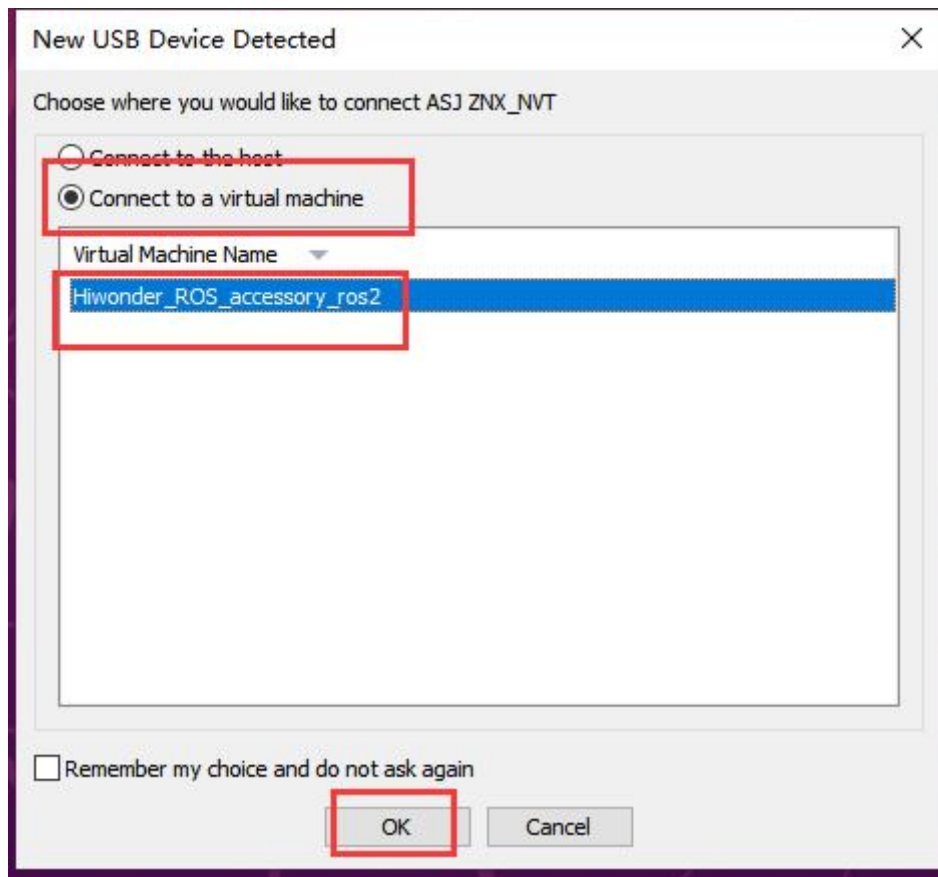


Usage of Depth Camera

Before starting, make sure that you have followed the "Lesson 1 Depth Camera Configuration" in the same directory as this lesson to set up the workspace. Ensure that you have successfully compiled the depth camera package based on "Lesson 2 Install Depth Camera ROS2 SDK".

1. Enable Camera Service

1) Attach the depth camera to the computer and establish its connection with the virtual machine.



2) Run the command to start the camera.

```
./run_ascamera_node.sh
```

```
hiwonder@ubuntu:~/ros_ws$ ./run_ascamera_node.sh
```

We offer two approaches for image visualization. One option involves utilizing `rqt_image_view` to observe 2D images, while the alternative is to employ `rviz`

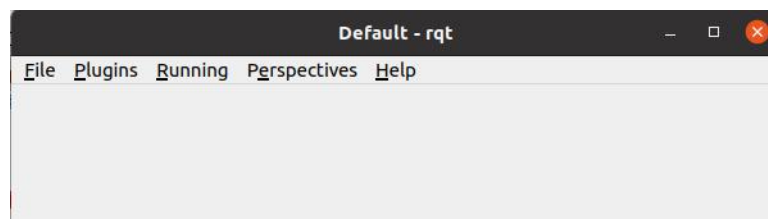
for visualizing 3D images. You are free to select the method that aligns with your particular needs. It is crucial to emphasize that during image viewing, the camera service should remain active and not be closed.

1.1 View Image Using rqt_image_view Tool

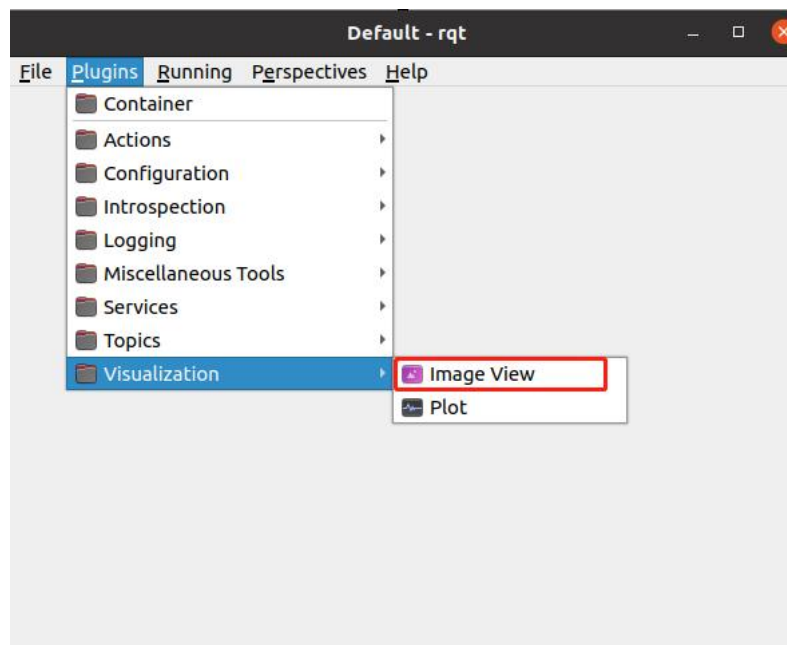
1) Open a new terminal, and execute the command to start the image viewing tool.

```
rqt  
hiwonder@ubuntu:~/Desktop$ rqt
```

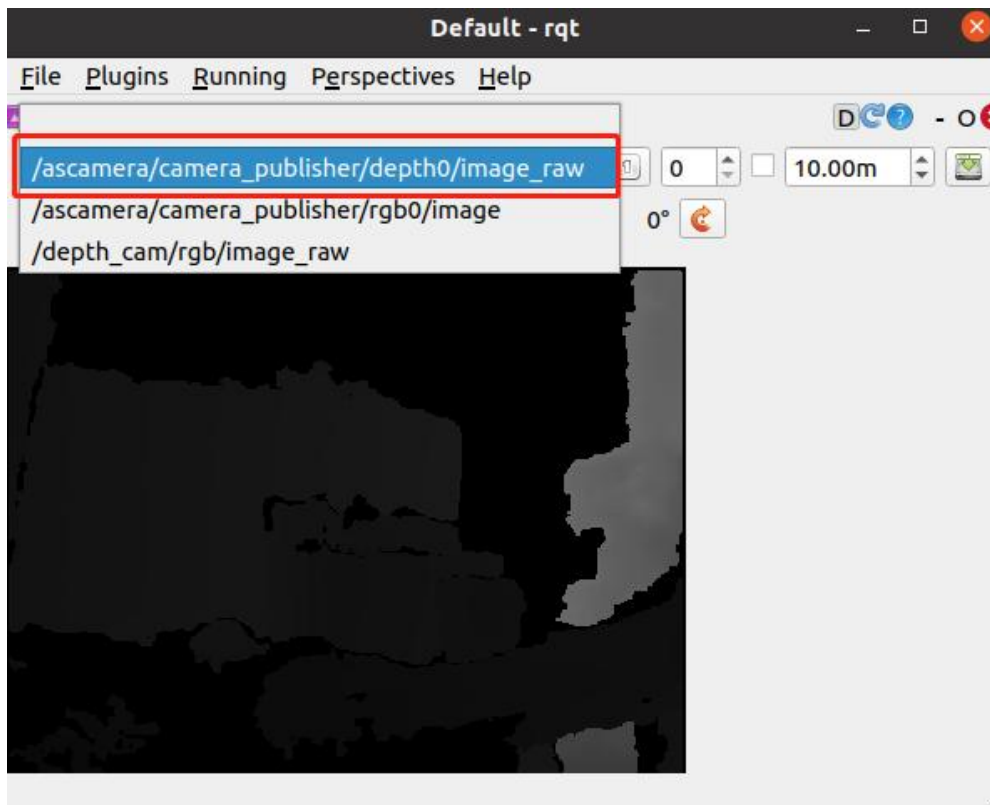
When the below interface appears, it means that the tool is started successfully.



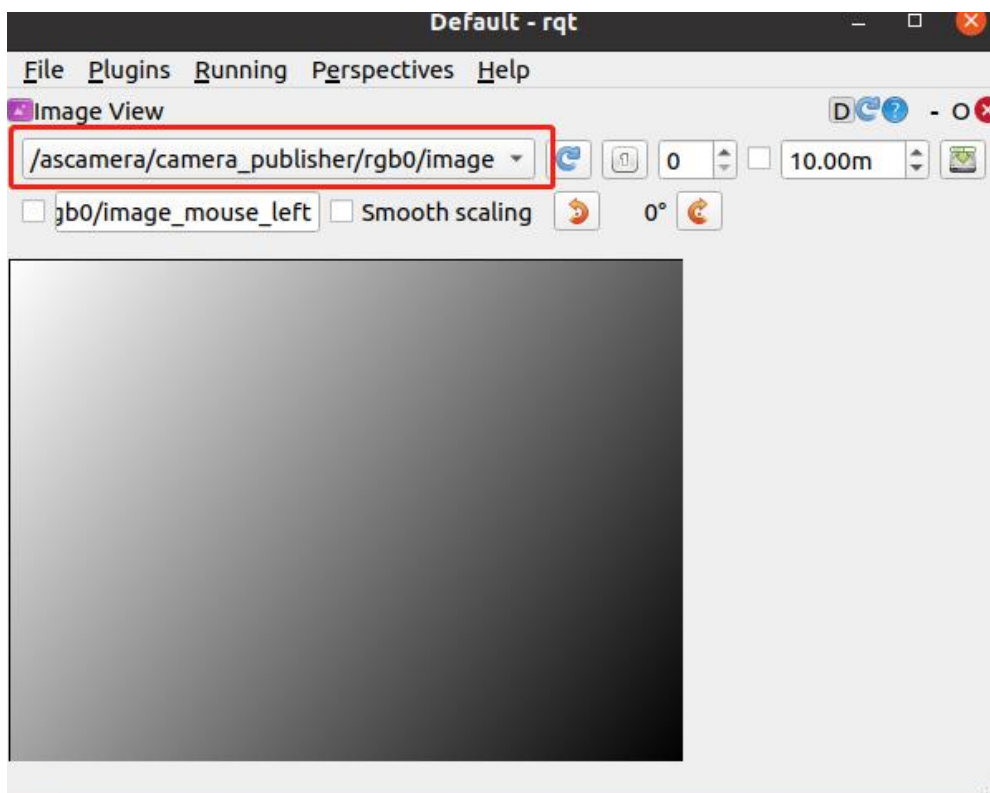
Select "Plugins > Visualization > Image View".



2) To view the infrared image, please select the corresponding topic.



3) When viewing the depth map, the red box position needs to be set as shown in the following figure:



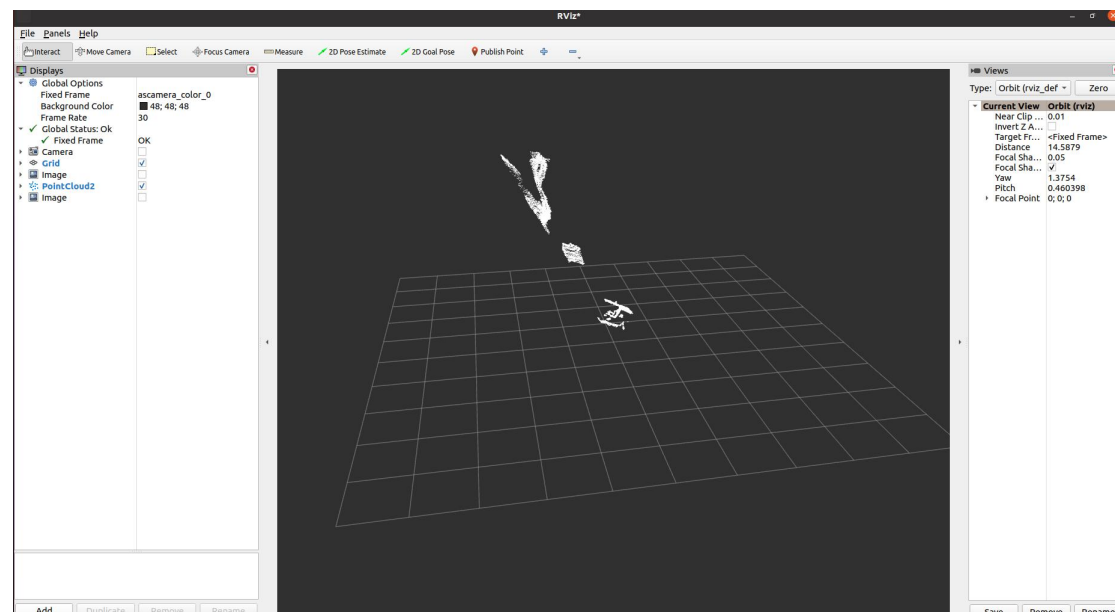
1.2 View Image Using rviz

1) Execute the command to initiate the tool.

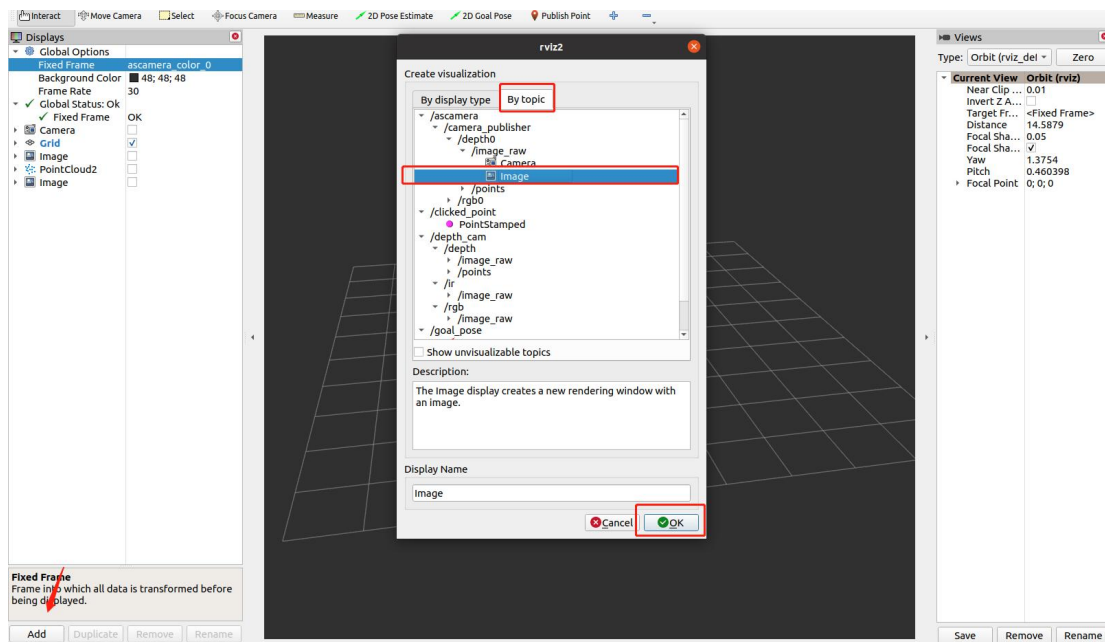
```
ros2 run rviz2 rviz2
```

```
hiwonder@ubuntu:~/Desktop$ ros2 run rviz2 rviz2
```

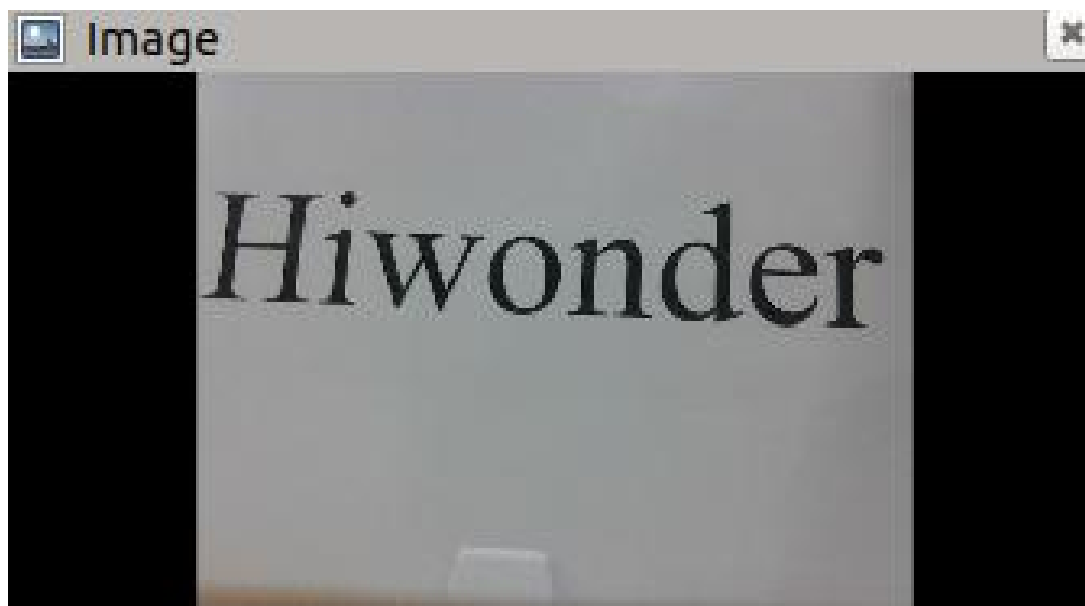
When the below interface appears, it means that the tool is started successfully.



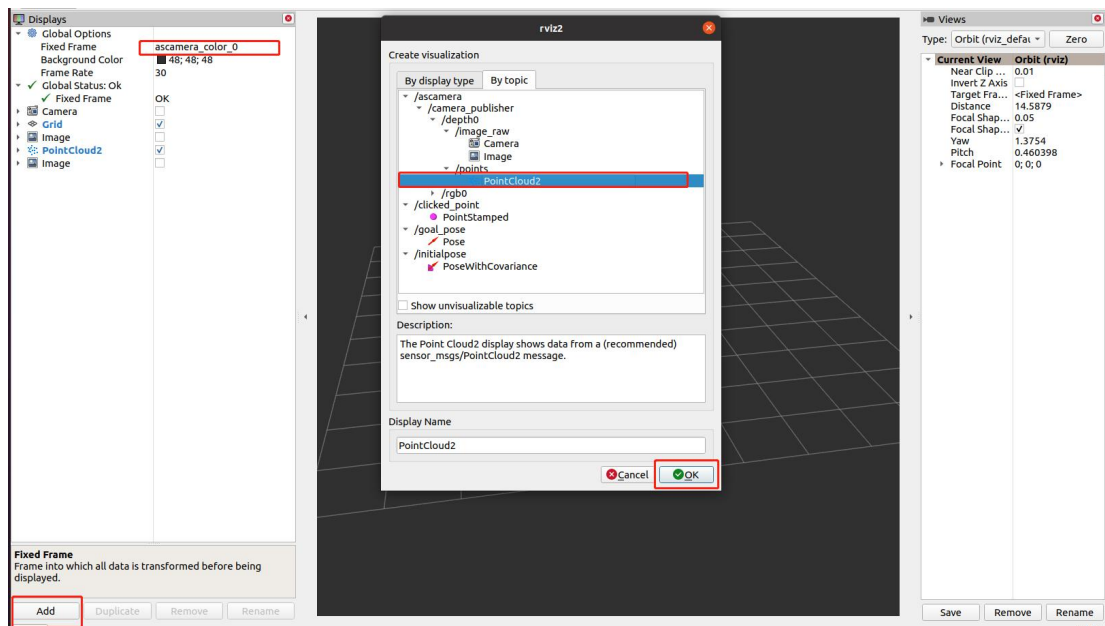
2) To access the camera feed, start by configuring the "Fixed Frame" to "camera_link." Then, click on "Add," select "By topic -> camera -> color -> Image_raw -> Image," and finally, click "OK."



When the interface below appears, it indicates that the tool has been successfully launched.



3) To view the depth map, set 'Fixed Frame' as 'camera_link', then click-on 'Add' to select 'By topic ->camera->depth->Image_raw->Image', and click-on 'OK'.



When the interface below appears, it indicates that the tool has been successfully launched.

